

Welcome to the minitutorial!

**To encourage interaction, please come to the ground floor
and site near the front row.**

<https://github.com/PerformanceEstimation/Tutorial-SIAM-OP26>



Performance Estimation Problems (PEPs)

Systematic analysis and design of first-order optimization algorithms
via the performance estimation framework

François Glineur, Adrien Taylor

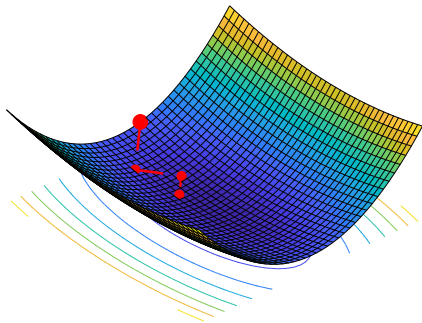


SIAM Conference on Optimization, tutorial session – June 2026



$$\min_{x \in \mathbb{R}^d} f(x)$$

→ usually solved via **iterative algorithm** generating sequence $x_0, x_1, \dots, x_N$.



Gradient descent (stepsize α)

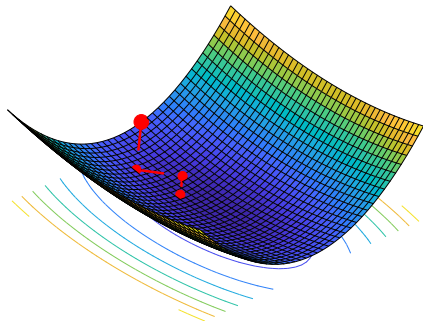
for $k = 0, 1, \dots$ **do**

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end for

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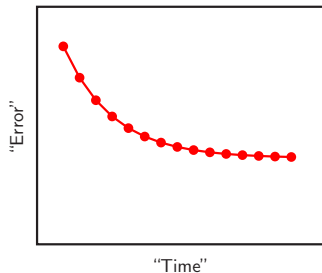
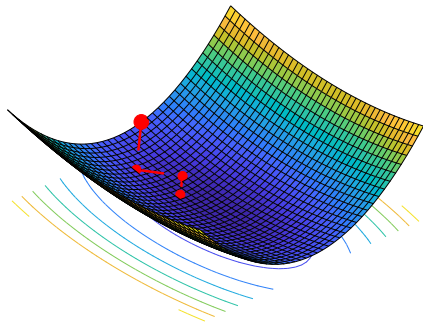
end for

What to expect from the output of the algorithm?

For instance: **bounds** on certain notions of “error”: $f(x_k) - f(x_*)$, $\|x_k - x_*\|$, $\|\nabla f(x_k)\|$, etc.

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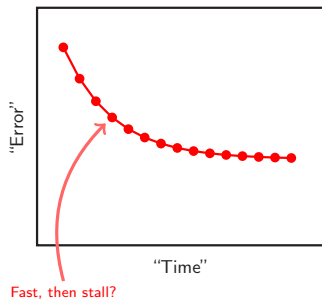
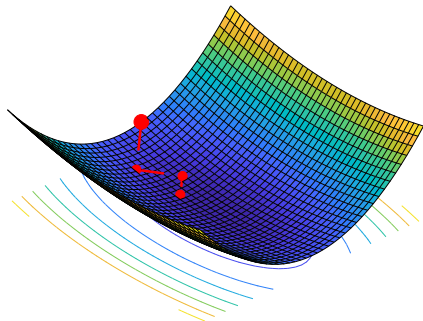


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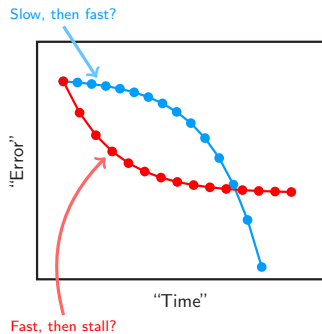
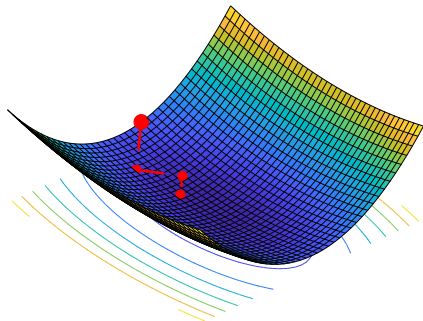


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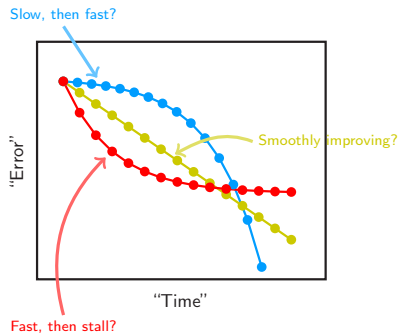
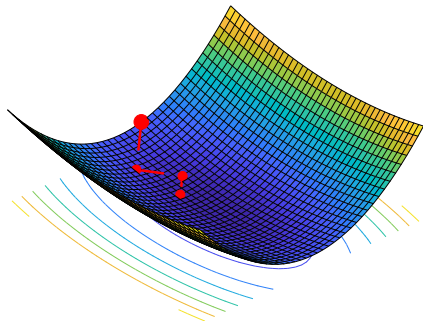


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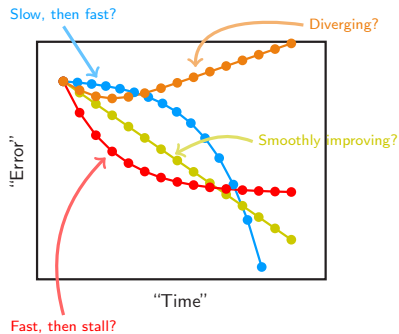
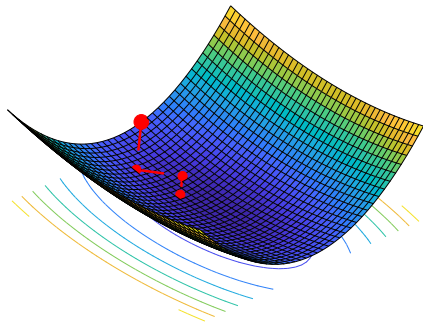


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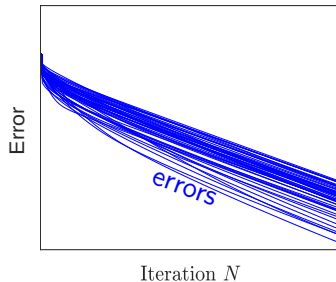
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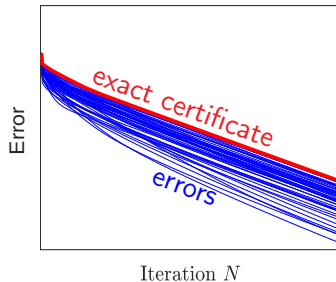
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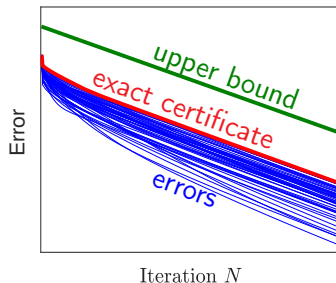
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Organization and learning outcomes (1/3)

What we would like you to learn with this tutorial:

- ◇ Base understanding of performance estimation problems (PEPs),
- ◇ leverage PEPs for numerical comparison of base first-order algorithms,
- ◇ leverage PEPs to obtain simple convergence proofs (numerics and symbolics),
- ◇ leverage PEPs for algorithm design (numerics and symbolics).

Tutorial material:

[https://github.com/PerformanceEstimation/
Tutorial-SIAM-OP26](https://github.com/PerformanceEstimation/Tutorial-SIAM-OP26)



Organization and learning outcomes (2/3)

Part 1 (now, Wed June 3, 9:15 AM - 10:45 AM):

- ◇ Talk: understand the basics of PEPs: context, problem statement, SDPs, interpolation
about 40mins
- ◇ Hands-on: numerical comparisons of FO methods using PEPs
about 30mins
- ◇ Talk: dual formulations and obtaining performance bounds with proofs
about 20mins

Organization and learning outcomes (3/3)

Part 2 (this PM, Wed June 3, 3:15 PM - 4:45 PM):

- ◇ Talk: proof structures
about 15mins
- ◇ Hands-on: numerical and symbolic tools for proving convergence rates
about 30mins
- ◇ Talk: designing first-order methods: principles and an example
about 10mins
- ◇ Hands-on: numerical and symbolic tools for designing a first-order method
about 20mins
- ◇ Discussions on going further
about 15mins

Ressources

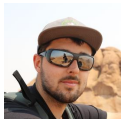
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Tools:

- ◇ Jupyter notebooks, or colab (here).
- ◇ Optimization tools/packages:
 - convex optimization interfacing/modeling tool: CVXPY.
 - Semidefinite (SDP) solvers: get academic MOSEK if possible (here).
- ◇ Symbolic computations:
 - SymPy [**OK for this minitutorial**], or consider Wolfram Engine (here+jupyter).
 - Mathematica is better (but expensive) — free (limited) version via cloud: here.

Please: do interrupt with questions!

Software/material built with many great colleagues, among which:



Baptiste
Goujaud



Céline
Moucher



Julien
Hendrickx



Aymeric
Dieuleveut



Shuvomoy
Das Gupta



Daniel
Berg Thomsen

Wooclap time!
(don't close it, you will need it later too!)



<https://app.wooclap.com/ZHBGHP?from=instruction-slide>